



INTRODUCTION TO SOFTWARE ENGINEERING

Overview of Course
(30 theory hours + 30 lab hours)

Ph.D Phan Phương Lan (pplan@ctu.edu.vn)

Software Engineering Department

College of Information and Communication Technology

Outline

- Course Learning Outcomes
- Course Content
- Textbook and References
- Assessment - Schedule

Course Learning Outcomes

Knowledge

CO1	Explain the concepts of software engineering, as well as the characteristics of each phase in the software development process.
CO2	Distinguish the software measurement and cost estimation methods.

Skills

CO3	Perform analysis, design, and unit testing for a specific software project with a selection of functions.
CO4	Collaborate in a team to complete a specific software development project.

Attitudes/Autonomy/Responsibilities

CO5	Complete assigned tasks and adhere to rules in software development.
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Course Content

- Part I: Overview of software engineering
- Part II: Software process
- Part III: Software measurement and cost estimation

Course Content

- Part I: Overview of software engineering
 - What is software engineering?
 - Software process models
 - Software project management
 - Project schedule
 - Project personnel
 - Quality management
 - Software configuration management
 - ...

Course Content

- Part II: Software process
 - Software specification
 - Design
 - Implementation
 - Validation (software testing)
 - Deployment
 - *Software maintenance*

Course Content

- Part III: Software Measurement and Estimation
 - Software metrics
 - Cost estimation

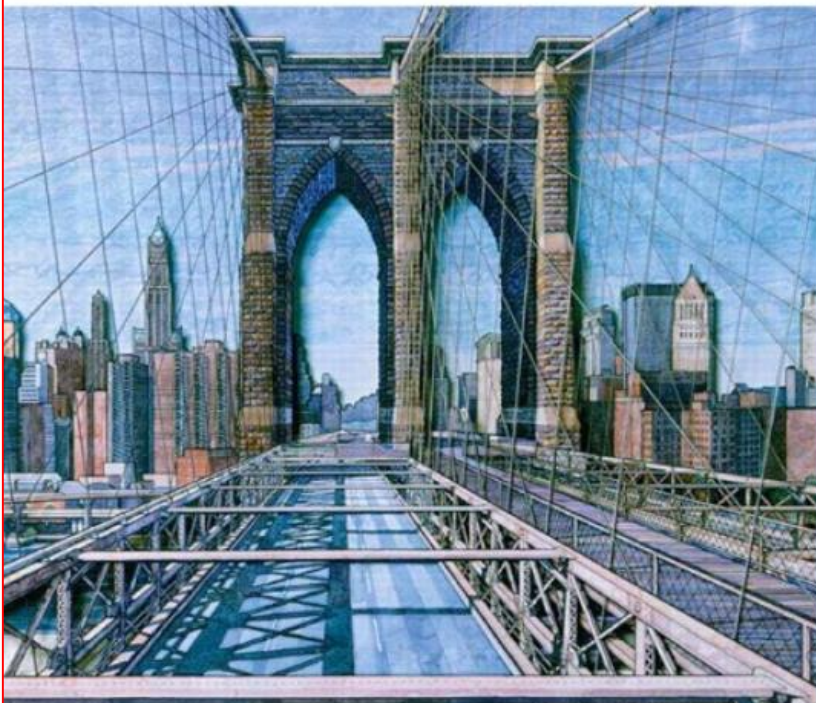
Textbook and References

- Class web: <https://elearning.ctu.edu.vn/course/view.php?id=1767>
- Textbooks:
 - Shari Lawrence Pfleeger, Joanne M. Atlee, Software Engineering theory and practice, Pearson Prentice Hall, 3th edition, 2006.
 - Ian Sommerville, Software Engineering, Addison – Wesley, 9th edition, 2009.
 - Hiep Xuan Huynh, Lan Phuong Phan, Introduction to Software Engineering, Can Tho University Publishing House, 2010.

Software Engineering

THEORY AND PRACTICE

THIRD EDITION



SHARI LAWRENCE PFLEEGER
JOANNE ATLEE

SOMMERVILLE

SOFTWARE ENGINEERING

9





TS. HUỖNH XUÂN HIỆP
Th.S PHAN PHƯƠNG LAN
Nhập Môn
CÔNG NGHỆ PHẦN MỀM

Giá:.....

TS. HUỖNH XUÂN HIỆP - Th.S PHAN PHƯƠNG LAN



Nhập Môn

CÔNG NGHỆ PHẦN MỀM

TS. HUỖNH XUÂN HIỆP
Th.S PHAN PHƯƠNG LAN

Nhập Môn **CÔNG NGHỆ PHẦN MỀM**

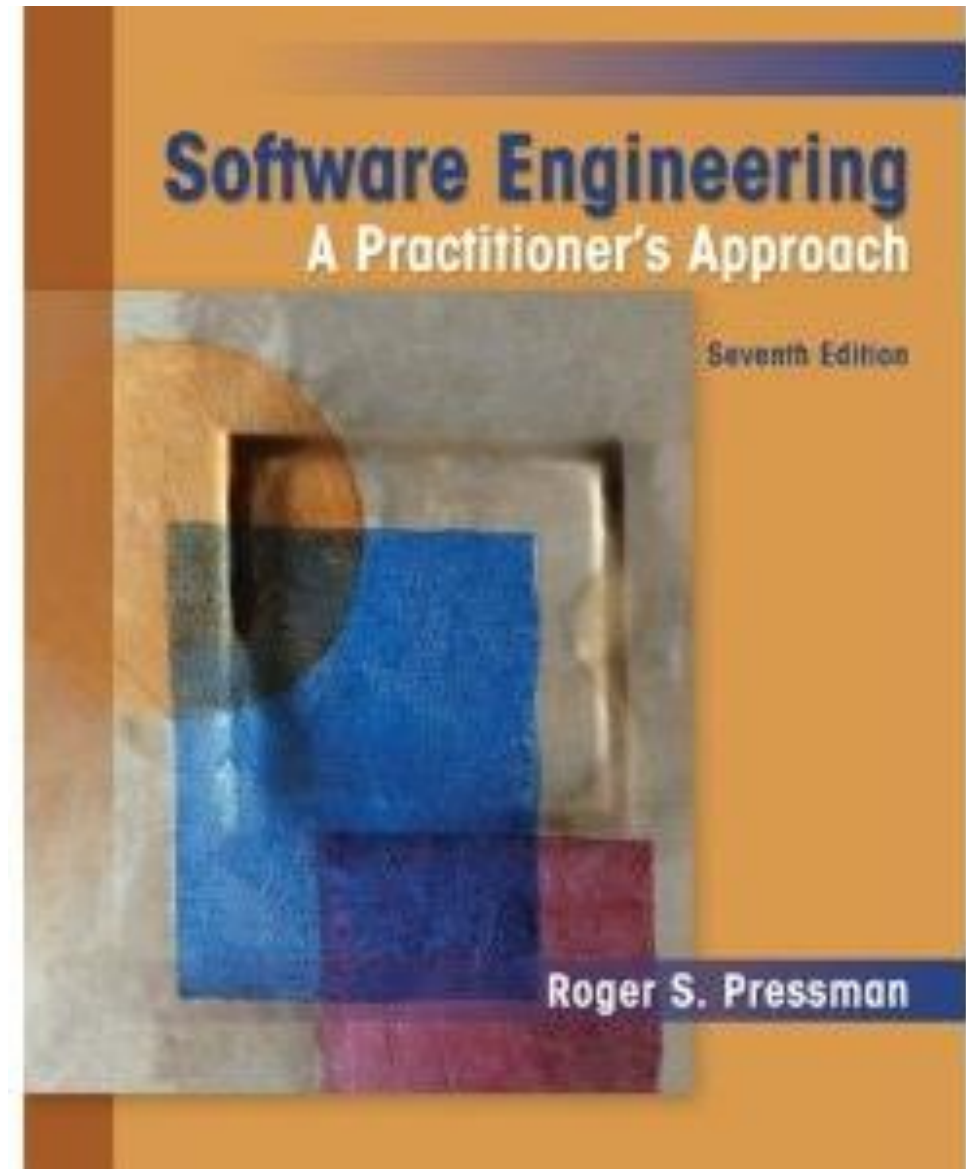


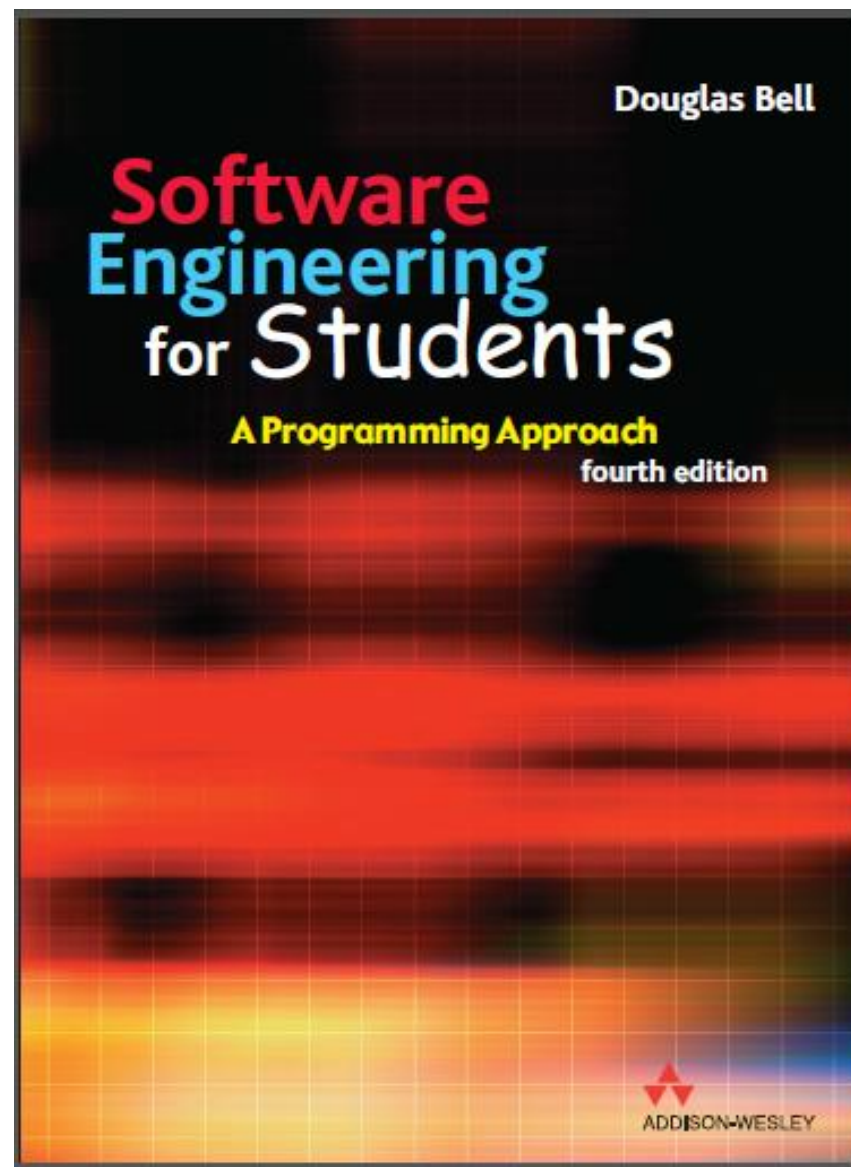
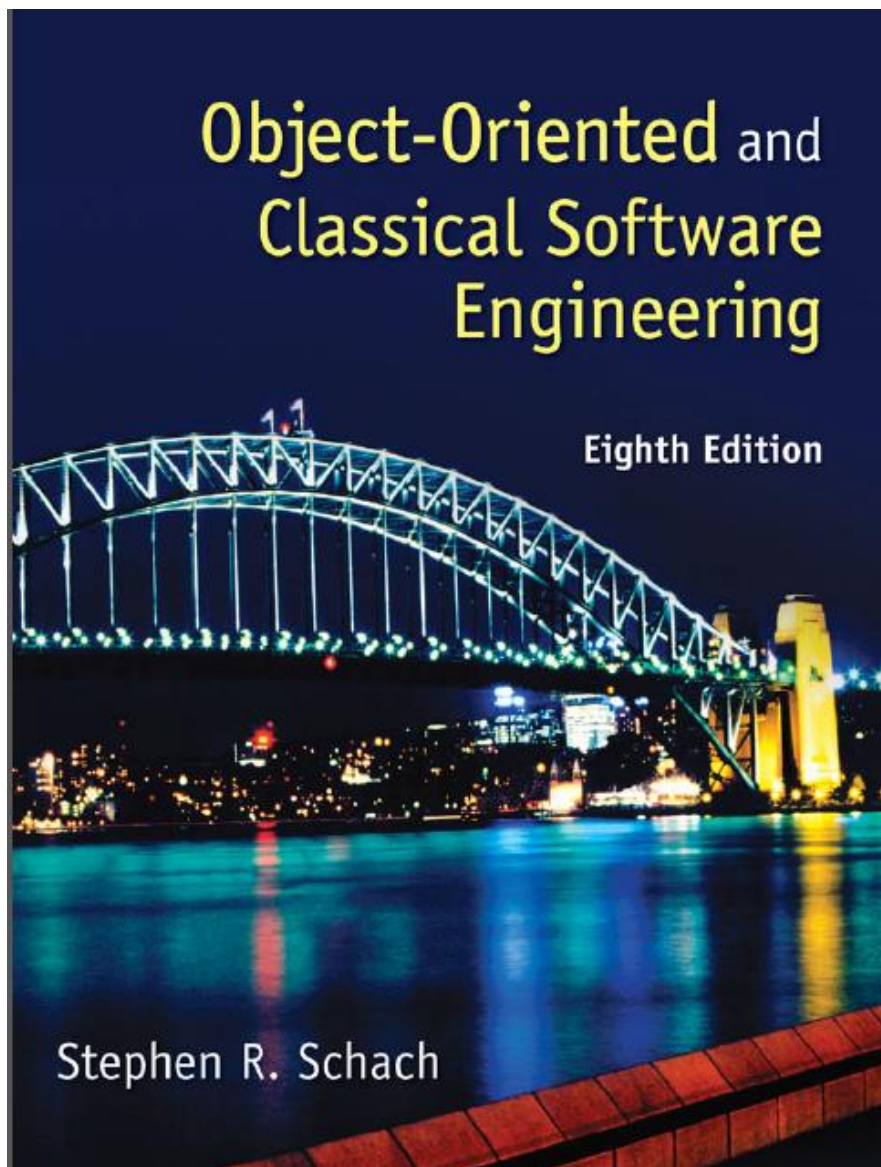
NHÀ XUẤT BẢN ĐẠI HỌC CẦN THƠ

Textbook and References

■ References:

- Hans Van Vliet, Software Engineering principles and practice, John Wiley, 2000.
- Pressman, Roger S., Software Engineering: A Practitioner's Approach, McGraw-Hill, 5th edition, 2003.
- ...





Assessment – Schedule

#	Assessment Method	Weight	Week	Course Learning Outcomes				
				CO1	CO2	CO3	CO4	CO5
1	Writing (Assignment: Write an essay on the topic of software failures)	5%	2					✓
2	Writing (Assignment: Estimate the cost of software development)	5%	10		✓			
3	Group Project (Develop a software application)	50%	2 – 13					
	- <i>Report</i>	35%	4 – 12	✓		✓	✓	✓
	- <i>Group presentation</i>	10%	13	✓		✓	✓	✓
	- <i>Group evaluation (Peer assessment)</i>	5%	5 – 13					✓
4	Final Exam	40%	12	✓	✓			
	Total	100%						

Assessment – Schedule

- A group of 3 - 4 students.
- Propose a software project to be developed.
- A set of final deliverables – including the plan, analysis, design, unit test documents and source code (if applicable) – must be submitted at the end of semester.

Q&A